

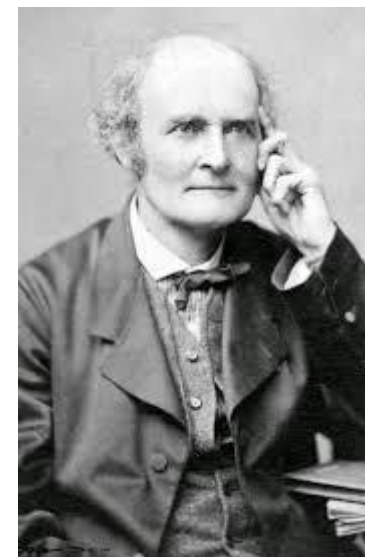
Math Primer

(9. Matrices)

Contents

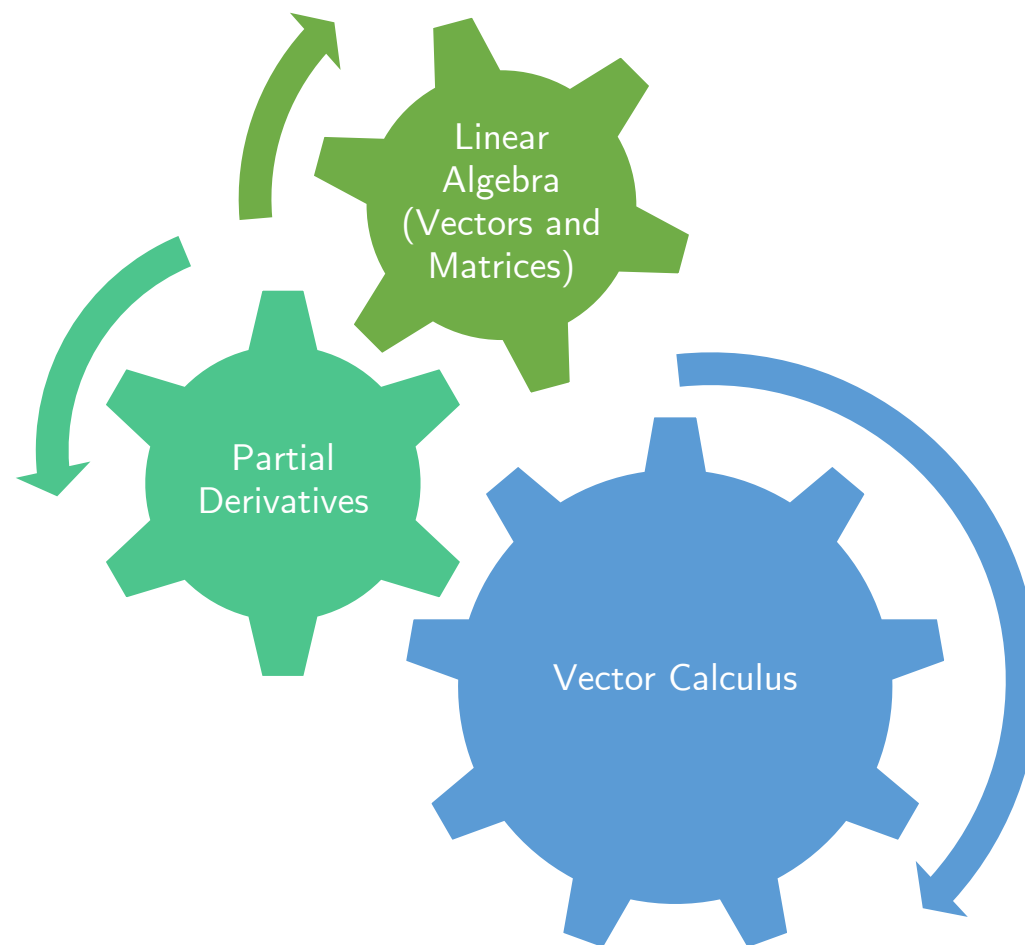


- ▶ Matrix representation
- ▶ Types of matrices
- ▶ Matrix Sum, Product, Transpose, Inverse
- ▶ Determinant, Minors and Cofactors
- ▶ Special Matrices
- ▶ Row Space, Column Space, Rank
- ▶ Transforming Vector spaces
- ▶ System of Linear Equations
- ▶ Eigen Values and Eigen Vectors



Arthur Cayley
(1821 – 1895)

The Multivariable Calculus Landscape



Matrices

- Think of Matrix as stacking of vectors

Fruit Basket 1



(3 apples, 4 Oranges)

Fruit Basket 2



(4 apples, 3 Oranges)

	Age	Income	...
ID1			
ID2			

→ Fruit Basket 1

→ Fruit Basket 2

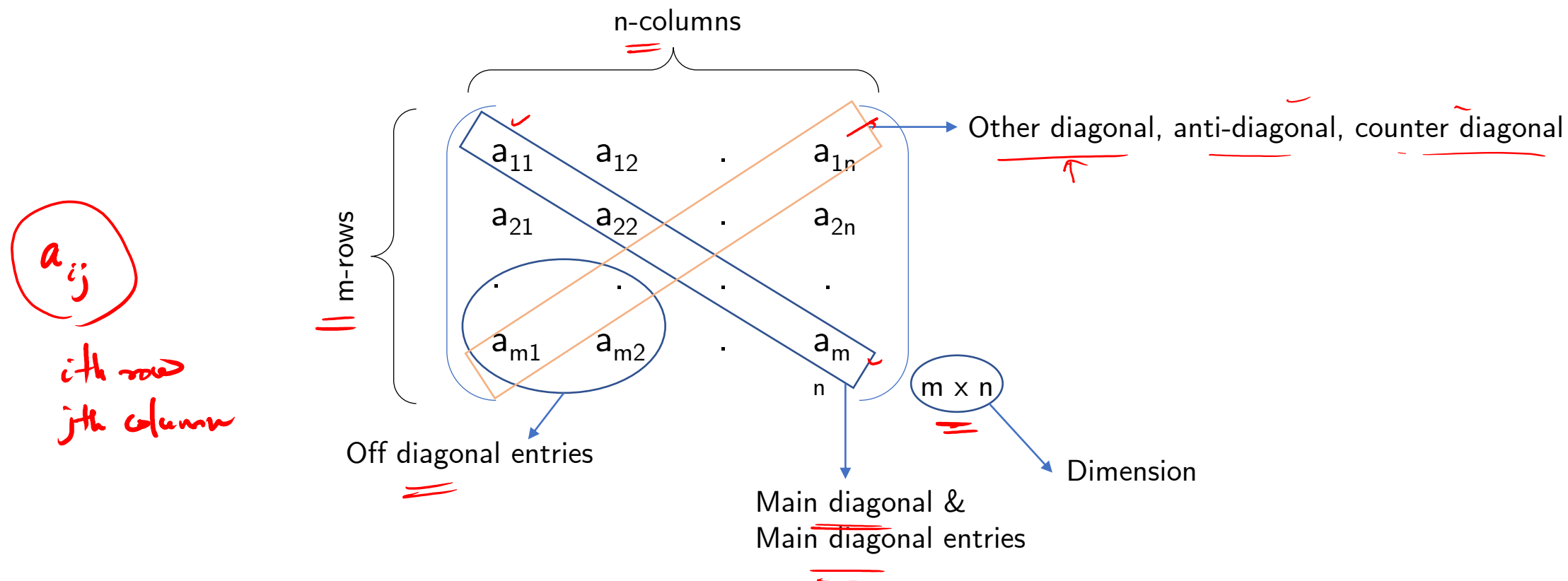
3	4
4	3

(Matrix)

- 2d matrices are most common where the need is to store data in rows and columns format. Typically, the columns remain static, the rows represent the instances or cases which are dynamic and added or deleted with ease. These are referred to as data frame in programming languages.

Matrices - Nomenclature

► Structure of a Matrix , Nomenclature...



Refer to a cell as a_{ij} . (stands for the element in ith row and jth column)

Matrices - Types

► Types of Matrices

$$\begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}_{2 \times 2}$$

Square Matrix

(no of Rows = no of columns)

$$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}_{2 \times 2}$$

Diagonal Matrix

(off-diagonal entries are 0)

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{2 \times 2}$$

Identity Matrix

($a_{ij} = 1$ if $i = j$, 0 otherwise)

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 1 & 3 \end{pmatrix}_{3 \times 3}$$

Symmetric Matrix

($a_{ij} = a_{ji}$)

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 5 & 0 \\ 3 & 1 & 3 \end{pmatrix}_{3 \times 3}$$

Lower Triangular Matrix

($a_{ij} = 0$ if $j > i$)

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}_{3 \times 3}$$

Upper Triangular Matrix

($a_{ij} = 0$ if $i > j$)

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}_{3 \times 3}$$

Null / Zero Matrix

($a_{ij} = 0$ everywhere)

Refer to https://en.wikipedia.org/wiki/List_of_named_matrices for more details

Matrices - SUM

- ▶ To sum 2 matrices, they need to be of equal size and the corresponding elements get added
- ▶ Output is also the same dimension
- ▶ $A + B = B + A$

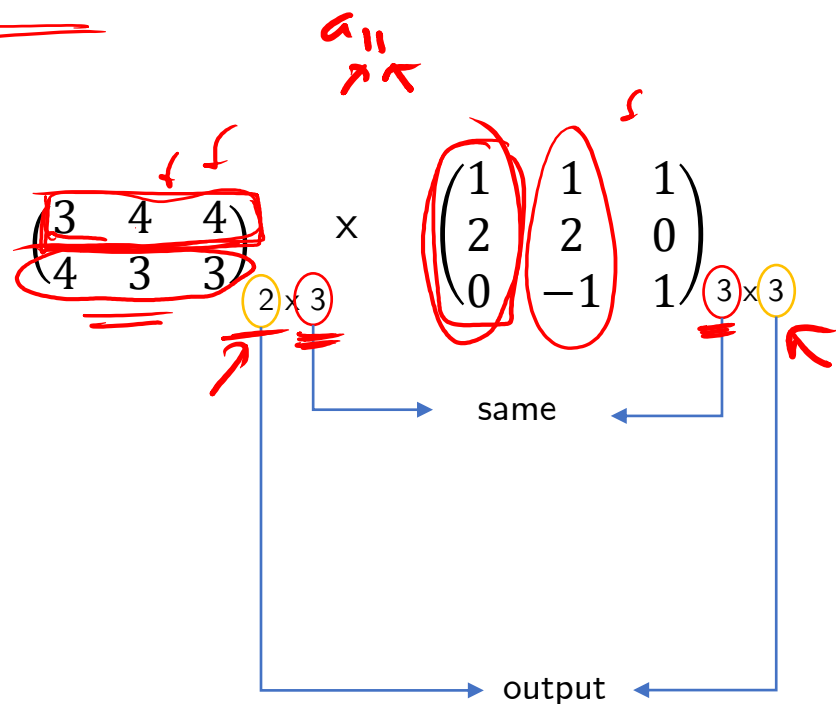
$$\begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}_{2 \times 2} + \begin{pmatrix} 0 & 1 \\ 2 & -2 \end{pmatrix}_{2 \times 2} = \begin{pmatrix} 3 & 5 \\ 6 & 1 \end{pmatrix}_{2 \times 2}$$

$$\begin{pmatrix} 3 & 4 \\ 4 & 3 \\ 4 & 3 \end{pmatrix}_{3 \times 2} + \begin{pmatrix} 0 & 4 \\ -2 & 1 \\ 3 & 1 \end{pmatrix}_{3 \times 2} = \begin{pmatrix} 3 & 8 \\ 2 & 4 \\ 7 & 4 \end{pmatrix}_{3 \times 2}$$

$$\begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}_{2 \times 2} + \begin{pmatrix} 0 & 4 \\ -2 & 1 \\ 3 & 1 \end{pmatrix}_{3 \times 2} \quad \times$$

Matrices - PRODUCT

- ▶ Two matrices, $A_{m \times n}$ and $B_{p \times q}$ can be multiplied if $n = p$, and the result will be of size $(m \times q)$
- ▶ In the resulting matrix, the cell (i, j) is the dot product of i th row of 1st matrix and j th column of 2nd matrix
- ▶ $AB \neq BA$ (not commutative)



$$\begin{aligned}
 &= \begin{pmatrix} \left(\begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right) & \left(\begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \right) & \left(\begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right) \\
 \left(\begin{pmatrix} 4 \\ 3 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right) & \left(\begin{pmatrix} 4 \\ 3 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \right) & \left(\begin{pmatrix} 4 \\ 3 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right) \end{pmatrix} \\
 &= \begin{pmatrix} 11 & 7 & 7 \\ 10 & 7 & 7 \end{pmatrix}
 \end{aligned}$$

Matrices - TRANSPOSE

- ▶ Matrix TRANSPOSE is denoted as A' or A^T . A transpose operation converts rows into columns and vice versa. Meaning the cell (i, j) in original matrix will appear in cell (j, i) of the transposed matrix.

$$A \rightarrow a_{ij}$$

$$A^T \rightarrow (ji)$$

$$A^T, A'$$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

$$A' = A^T = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

- ▶ Properties of Transpose

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(kA)^T = kA^T$$

$$(AB)^T = B^T A^T$$

$$k \rightarrow \text{scalar } 5$$

$$(k)^T = k$$

$$(5)_{1 \times 1} \rightarrow (5)_{1 \times 1}$$

Matrices – DETERMINANT, MINORS, COFACTORS

$$\underline{|A|} = \underline{\begin{vmatrix} a & b \\ c & d \end{vmatrix}} = \underline{ad - bc} \quad (\text{Determinant of a } 2 \times 2 \text{ matrix})$$

$$\underline{|A|} = \underline{\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}} = \underline{a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}} = \underline{aei + bfg + cdh - ceg - bdi - afh}$$

$$- \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}^{M_{22}} = e \begin{vmatrix} a & c \\ g & i \end{vmatrix}$$

minors

cofactors

M_{11} M_{12} M_{13} M_{ij} (for cell (i,j))

$(-1)^{1+1}M_{11}$ $(-1)^{1+2}M_{11}$ $(-1)^{1+3}M_{11}$ $(-1)^{i+j}M_{11}$

$(-1)^{i+j}M_{ij}$ $i=1, j=1$ $i=1, j=2$

- ▶ $\underline{\det(A)}$ or $\underline{|A|} =$ row-wise or column-wise sum-product of elements and their co-factors
- ▶ $\underline{adj(A)} =$ transpose of cofactor matrix
- ▶ $\underline{\det(AB)} = \underline{\det(A) \det(B)}$
- ▶ $\underline{\det(A^T)} = \underline{\det(A)}$
- ▶ $\underline{\det(A^{-1})} = \underline{1/\det(A)}$

$$\begin{pmatrix} cf_{11} & cf_{12} & cf_{13} \\ - & - & - \\ - & - & - \end{pmatrix}^T$$

Matrices – DETERMINANT, MINORS, COFACTORS



Exercise – Verify the adjoint of the matrix below.

$$\text{adj} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{bmatrix}$$

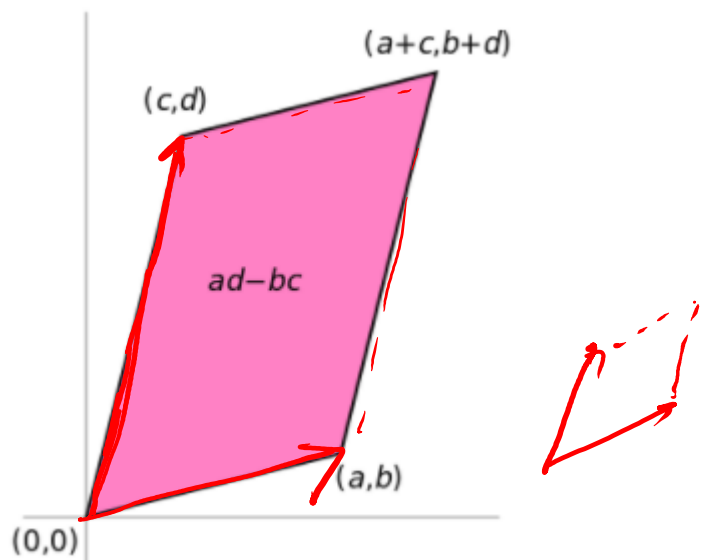
$$\text{Adj}(A) = [\text{cofactors}]^T$$

$$\begin{bmatrix} -3 & 6 & -3 \\ - & - & - \\ - & - & - \end{bmatrix}^T \quad \left[\right.$$

Matrices – DETERMINANT (Geometric Interpretation)

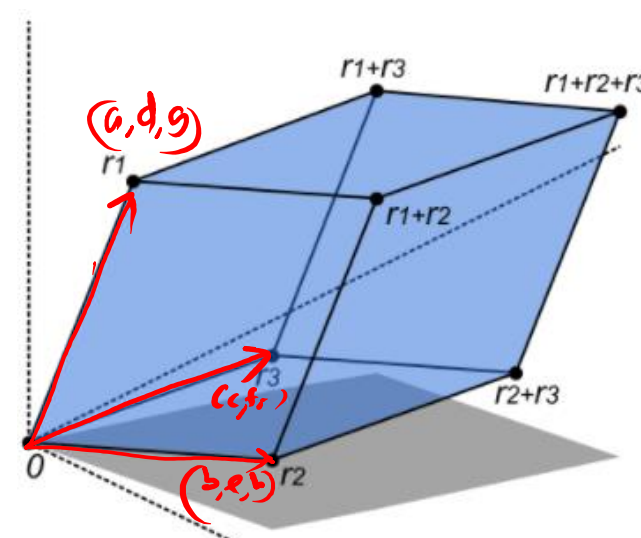


- Determinant is a scalar quantity. The geometric interpretation of determinant is that it captures the volume enclosed by the vectors



$$|A| = \begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc$$

Area of the parallelogram



$$|A| = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

Volume of a parallelepiped

Matrices – INVERSE

- ▶ For a matrix A to be invertible, it must be a square matrix and $\det(A) \neq 0$
- ▶ Inverse is given using adjoint and determinant : $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$

Properties

$$\underline{AA^{-1}} = \underline{A^{-1}A} = \underline{I}$$

$$\underline{(kA)^{-1}} = \underline{\left(\frac{1}{k}\right)A^{-1}}$$

$$\underline{(AB)^{-1}} = \underline{B^{-1}A^{-1}}$$

$$\underline{(A^T)^{-1}} = \underline{(A^{-1})^T}$$

$$\begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}$$

$$(A^{-1})^{-1}$$

$$= A$$

$$(5)^{-1} = \frac{1}{5}$$

$$A = \begin{pmatrix} 1 & 2 & 0.5 \\ 0.5 & -1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} 5 & 1 & -4.5 \\ -1.5 & -0.5 & 1.75 \\ -2 & 0 & 2 \end{pmatrix}$$

Matrices – Special types

► Here are some special matrices ...



Condition	Name
$A^T = A$	Symmetric ←
$A^T = -A$	Skew-symmetric
$A^T = A^{-1}$	Orthogonal
$A^n = A$	Idempotent
$ A = 0$	Singular
$\underline{x^T A x} > 0$ for all $x \neq 0$	Positive definite ✓
$\underline{x^T A x} \geq 0$ for all $x \neq 0$	Positive semi-definite ✓

Matrices – Column space, Row Space and Rank

- ▶ Recall that in a set of vectors, if no vector can be created by a linear combination of others, then they are called linearly independent.
- ▶ Another way to verify the above is if there exists a linear combination of all the vectors (where all the scalars are not 0) that leads to 0 , e.g., $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ are dependent, because

$$\overset{v_3}{\boxed{\begin{pmatrix} 3 \\ 5 \end{pmatrix}}} = \overset{\alpha}{(1)} \overset{v_1}{\begin{pmatrix} 1 \\ 2 \end{pmatrix}} + \overset{\beta}{(1)} \overset{v_2}{\begin{pmatrix} 2 \\ 3 \end{pmatrix}} \quad \text{or} \quad (1)\begin{pmatrix} 3 \\ 5 \end{pmatrix} - (1)\begin{pmatrix} 1 \\ 2 \end{pmatrix} - (1)\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$


- ▶ Column space is the span of column vectors
- ▶ Row space is the span of row vectors (= column space of the transpose)
- ▶ The dimensions spanned by the column vectors is called 'rank' of a matrix (= number of linearly independent columns). If all column vectors are independent, matrix has **full rank**.
- ▶ Row space = Column space for a square matrix
- ▶ Rank of a square matrix = number of linearly independent columns or rows
- ▶ If a square matrix is NOT full rank, then its determinant = 0 (*singular*)

Matrices – Column space, Row Space and Rank (Examples)



	<u>Column space</u> <u>whole of $\mathbb{R}^2$</u>	<u>Row space</u> <u>whole of $\mathbb{R}^2$</u>	<u>Rank</u> <u>2 (full rank)</u>
	<u>whole of $\mathbb{R}^2$</u>	<u>2D plane in $\mathbb{R}^3$</u>	<u>2 (Full row rank)</u>
	<u>A line in $\mathbb{R}^2$</u>	<u>A line in $\mathbb{R}^2$</u>	<u>1</u>
	<u>2D plane in $\mathbb{R}^3$</u>	<u>2D plane in $\mathbb{R}^3$</u>	<u>2</u>
	<u>A point in $\mathbb{R}^2$</u>	<u>A point in $\mathbb{R}^2$</u>	<u>0</u>
	<u>Whole of $\mathbb{R}^3$</u>	<u>Whole of $\mathbb{R}^3$</u>	<u>3 (full rank)</u>
	<u>A line in $\mathbb{R}^2$</u>	<u>A line in $\mathbb{R}^3$</u>	<u>1</u>
	<u>Whole of $\mathbb{R}^2$</u>	<u>A 2D-plane in $\mathbb{R}^4$</u>	<u>2 (Full row rank)</u>

Matrices – Multiplying a Matrix with a Vector

- When we multiply a matrix with a vector, we get a transformed vector. It's convenient to think of transforming vector spaces instead of just a single vector. That way we can classify some very useful matrices based on their resulting transformation of vector spaces such as Rotation, Stretching, Shear, Projection etc....

$A \cdot x = b$
 $\begin{matrix} m \times n & n \times 1 & \\ \hline & m \times 1 \end{matrix}$
 $b = x$

$Ax = b$
 $\begin{matrix} \downarrow & \downarrow & \downarrow \\ \text{(Matrix)} & \text{(Vector)} & \text{(Transformed Vector)} \end{matrix}$

$A = I$
 $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$
 $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ a line

$A = \begin{pmatrix} 5 & 2 \\ 1 & 3 \end{pmatrix}$ $x = \begin{pmatrix} x \\ y \end{pmatrix}$ $b = \begin{pmatrix} 5 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

Row Picture (dot product)
 Column Picture (Column space)
 (**b** must be in **A**'s column space)

$b = \begin{pmatrix} 5x + 2y \\ x + 3y \end{pmatrix}$
 $b = x \begin{pmatrix} 5 \\ 1 \end{pmatrix} + y \begin{pmatrix} 2 \\ 3 \end{pmatrix}$
 $\begin{pmatrix} 5x \\ x \end{pmatrix} + \begin{pmatrix} 2y \\ 3y \end{pmatrix}$

Matrices – Solving a system of linear equations

- A system of linear equations can be solved using Matrix Algebra. Say we have 2 equations with 2 unknowns

$$\begin{cases} 5x + 2y = 9 \\ x + 3y = 7 \end{cases}$$

$$\begin{matrix} \cancel{A}x = \cancel{b} \\ A \quad x \quad b \\ \begin{pmatrix} 5 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 9 \\ 7 \end{pmatrix} \end{matrix}$$

$$Ax = b$$

Multiply A^{-1} on both sides to get

$$\begin{aligned} A^{-1}Ax &= A^{-1}b \\ Ix &= A^{-1}b \\ x &= A^{-1}b \end{aligned}$$

$$IA = A$$

$$A^{-1} = \begin{pmatrix} 0.2308 & -0.1538 \\ -0.0769 & 0.3846 \end{pmatrix}$$

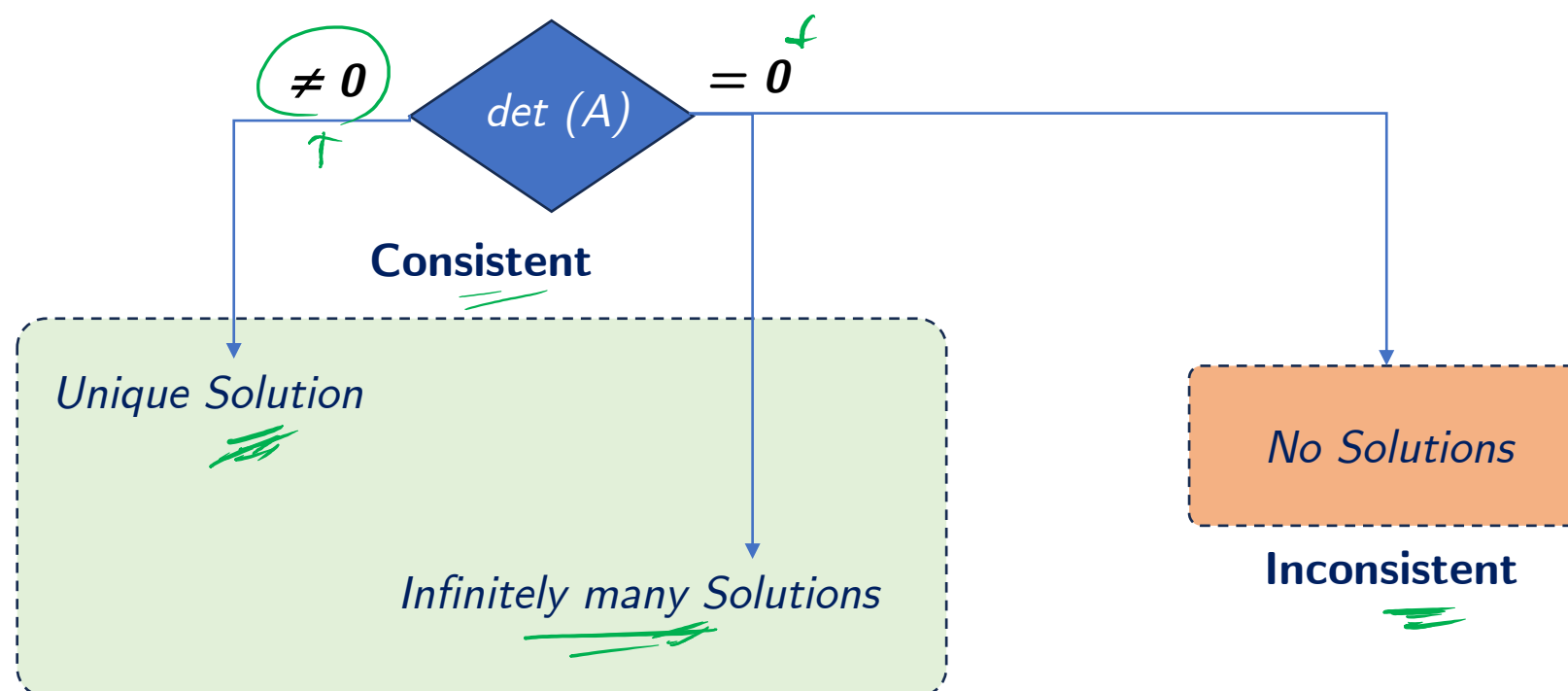
$$|A| = 13 \neq 0$$

$$\text{Solution : } x = \begin{pmatrix} 0.2308 & -0.1538 \\ -0.0769 & 0.3846 \end{pmatrix} \begin{pmatrix} 9 \\ 7 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

if A^{-1} exists ($\det(A) \neq 0$), solution exists for all b
 $\det(A) \neq 0$ means matrix is *full rank*

Matrices – Solving a system of linear equations

- ▶ A system of linear equations is called consistent if there is at least one solution and inconsistent if there are no solutions
- ▶ If $\det(A) = 0$, then the system is either consistent with infinitely many solutions or inconsistent (no solutions)



Matrices – Solving a system of linear equations

Exercise – Find the solution to the following system of equations.

a) $\begin{cases} x + 2y = 8 \\ 2x + 4y = 16 \end{cases}$ } consistent infinitely many

$$\begin{array}{r} \frac{1x + 2y = 8}{2x + 4y = 16} \rightarrow \begin{array}{l} x = a \\ y = 2 - a \end{array} \\ \hline \end{array}$$

$x = \begin{pmatrix} a \\ 2 - a \end{pmatrix}$, a – free parameter

b) $\begin{cases} x + 2y = 8 \\ 2x + 4y = 15 \end{cases}$ } inconsistent no solⁿ.

$$\left| \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \right| = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$= \rightarrow 2 - 2 = 0$$

$$\begin{array}{r} \frac{x + y = 2}{2x + 2y = 3} \text{ (inconsistent)} \\ \hline \end{array}$$

$$\left| \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \right| = 0$$

Matrices – Eigen Value and Eigen Vector

- ▶ We saw that matrix multiplied with a vector performs a linear transformation within the column space of the matrix.
- ▶ The vector(s) which only get stretched (scaled) and do not change direction is(are) called 'eigen vector(s)' of the matrix.
- ▶ The scalar by which each eigen vector gets stretched is the corresponding eigen value.
- ▶ Mathematically,

Handwritten notes:

A (circled) with arrows pointing to $v = ??$ and $\lambda = ??$

(λ, v) (circled) with an arrow pointing to A

$Av = \lambda v$

$A(2v) = \lambda(2v)$

Diagram illustrating the eigen equation:

$$Av = \lambda v$$

Labels with arrows:

- A is labeled "Matrix"
- v is labeled "Eigen Vector"
- λ is labeled "Eigen value"

Characteristic Polynomial

$\rightarrow \underline{Av = \lambda v}$, $\lambda \rightarrow \text{scalar}$
 $\underline{Av = \lambda Iv}$, $I \rightarrow \text{identity}$
 $\rightarrow \underline{(A - \lambda I)v = 0}$ $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ (Homogeneous equation), $\underline{v = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}}$

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{2 \times 2}$, $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}_{3 \times 3}$

- ▶ This system always has a solution $v = 0$ (trivial solution)
- ▶ If the above solution is not unique, there will be infinitely many solutions
- ▶ We are interested to see the structure of these infinite solutions

But for the system to have infinitely many solutions we need $|(A - \lambda I)| = 0$

$$\underline{|(A - \lambda I)| = 0}$$

Characteristic equation of A

Eigen Value and Eigen Vector - Example

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad \underline{A\mathbf{v} = \lambda\mathbf{v}} \quad \underline{(A - \lambda I)\mathbf{v} = \mathbf{0}}$$

$$\underline{|(A - \lambda I)| = 0} \quad \text{(Characteristic equation of A)}$$

$$(A - \lambda I) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix}$$

$$\underline{|(A - \lambda I)| = (2 - \lambda)^2 - 1 = 0}$$

$$\underline{\lambda^2 - 4\lambda + 3 = 0}$$

$$\lambda = 1, 3 \quad \text{(Eigen values)}$$

$$\lambda = 1$$

$$\lambda = 3$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{matrix} -x + y = 0 \\ x - y = 0 \end{matrix} \quad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{matrix} x + y = 0 \\ x + y = 0 \end{matrix}$$

$$\begin{pmatrix} 2 \\ -2 \end{pmatrix} \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Properties of Eigen Vectors and Eigen Values

- 99
- Trace of A (sum of diagonal entries) = sum of eigen values $\text{tr}(A) = \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i = \lambda_1 + \lambda_2 + \dots + \lambda_n$
 - Determinant of A = product of eigen values $\det(A) = \prod_{i=1}^n \lambda_i = \lambda_1 \lambda_2 \dots \lambda_n$
 - Eigen values of A^k = k^{th} power of eigen values $\lambda_1^k, \dots, \lambda_n^k$
 - A is invertible only when all eigen values are non-zero
 - Eigen values of A^{-1} are reciprocal of eigen values of A $\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$
 - Eigen vectors of a symmetric matrix are orthogonal
 - Eigen values of a positive definite matrix are all > 0
 - Eigen values of a positive semi-definite matrix are ≥ 0

$(1, 9)$
 (A^{-1}) $(1, \frac{1}{3})$

$x^T A x > 0$